25900242

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTS200 – (ESSENTIALS IN BIOLOGICAL TECHNIQUES (2024 ADMISSION ONWARDS)

Part A

Fill in the blanks

Answer all questions

Each question carries 1 marks

1. The first step in bioanalysis is _____ collection. [K] / [CO1]
   Ans; - Sample collection
2. Bioanalytical techniques are essential for _____ analysis of drugs and biomolecules. [K] / [CO1]
   Ans; - Quantitative
3. One advantage of the Ag/AgCl electrode is its _____. [K] / [CO2]
   Ans; - simple construction / low cost
4. The Beer–Lambert's law is also known as the law of _____. [K] / [CO2].
   Ans; - absorbance
5. One limitation of a _____ is its limited wavelength range. [K] / [CO2]
   Ans; - colorimeter
6. The principle of electrophoresis is based on the migration of charged particles when an ____is applied**.**
   Ans; - Electric field

7. ____in the electrophoresis buffer acts as a chelating agent that can bind cofactors of DNA ase enzymes. [U] / [CO2]
   Ans; - EDTA
8. The process of breaking cells to release cell organelle before centrifugation is called ___ [U] / [CO2]
   Ans; - Homogenization

9. …….....radiation of low energy can be measured by Liquid scintillation counter. [U] / [CO4]
   Ans: Beta
10.…….....is used to diagnose bone diseases. [U] / [CO4]
   Ans:Tc-99 or Sr-85

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

11.Define colorimeter. [K] / [CO2]

Ans;

- The colorimetry is commonly used analytical technique involved in quantitative estimation of color i.e. it is utilized to find out concentration of the colored substance in the sample solution e.g. water, biological samples at visible spectrum of light.
- A substance must be colorful or should have property of forming chromogens through the addition of reagents which will absorb light according to their color intensity to be measured.

12. Define ion-selective electrodes and give an example. [K] / [CO2]

Ans; -

Ion selective electrode (ISE) is an analytical technique used to determine the activity of ions in aqueous solution by measuring the electrical potential. . An example is a pH-sensitive glass electrode

13. Define pH and write its formula. [K] / [CO2]

Ans; -

pH is a measure of the acidity or basicity of a solution. pH can be defined as the –log of the H^+ ion concentration

14. State the advantages of a double cell over a single cell colorimeter [K] / [CO2]

Ans; -

Double cell Colorimeter	*Single cell Colorimeter*
Uses a beam splitter to divide the light from a single source into two paths. One beam passes through the sample, and the other passes through a reference blank at the same time.	A single beam of light passes through the reference blank and the sample one after another. The operator must manually swap the cuvettes and take separate readings.

15. Give two practical uses of Beer–Lambert's law. [K] / [CO2]

Ans; -

Students can write any 2 applications that they studied.

16. Explain the significance of using Beta-mercaptoethanol in SDS -PAGE? [K] / [CO2, CO3].

Ans;-

The β mercaptoethanol acts as a denaturing agent by reducing all disulphide bonds. This, along with SDS (sodium dodecyl sulfate), fully denatures the protein, ensuring it unfolds into a linear shape and can be separated solely based on its molecular weight

17. Cite any 2 advantages of TLC over paper chromatography? [U] / [CO2]

Ans; -

Students can write any 2 advantages that they studied

18. Comment on any 2 application of ion exchange chromatography. [U] / [CO2, CO3]

Ans; -

Students can write any 2 applications that they studied

19. Mention any 2 application of SEC in protein purification ? [K] / [CO2]

Ans; -

Students can write any 2 applications that they studied

20. Cite the importance of centrifugal force in seperating cellular components . [U] / [CO2]

Ans; -

- Centrifugal force is an intriguing apparent force that acts outwardly away from the centre of rotation. In short, particles in suspension experience a radial centrifugal force that moves them away from the axis of rotation.
- When objects are on a circular path, they experience what is known as centrifugal force, which gives the sensation of being pushed away from the centre.
- Under centrifugal force, denser particles migrate toward the bottom of the tube to eventually form a pellet; the lighter particles will remain in the supernatant.

21. What is radioimmunoassay? [U] / [CO4]

Ans: Radioimmunoassay (RIA) is a technique used to measure the concentration of an antigen, such as a hormone in a sample . It works by using a radioactive-labeled antigen and a specific antibody to compete for binding with the unlabeled antigen from the sample. The more unlabeled antigen is in the sample, the less radioactive-labeled antigen binds, allowing for a quantifiable measurement of the unknown sample's antigen concentration.

22. Explain the principle of incorporation of radioisotopes into biological tissues and cells.

- **Distribution**: The labeled compound is administered to the organism by injecting into the body thus entering cells, tissues, and organs.
- **Tracking**: Specialized detectors are used to monitor the radiation emitted by the radioisotope. This allows researchers to determine the distribution and localisation of the compound in the body.

Part C

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

23. Explain the applications of I-131 in medicine. [U] / [CO4]

24. Explain the major parts of a spectrophotometer and illustrate them with a diagram. [U] / [CO2]

Ans; -

INSTRUMENTATION

The general instrument set up for any spectrophotometry requires;

- *Radiation source* - It can be of Argon, Xenon lamb , Nernst glower etc..
- *Condensing lens* – Inorder to generate a parallel beam of light.
- *Monochromator* - to break the polychromatic radiation into component wavelength (or) bands of wavelengths.
- *Wavelength selection (slit)* – It can be achieved by absorption filters ,or by inference filters. Fused silica or quartz prism is used in UV regions
- *Sample holder* – Type of sample holder depends on the transmission range of the materials. Generally glass is used in UV regions (300-500nm)
- *Photosensitive detector* and an associated readout system -Detectors generate electronic signals which are proportional to the transmitter light.

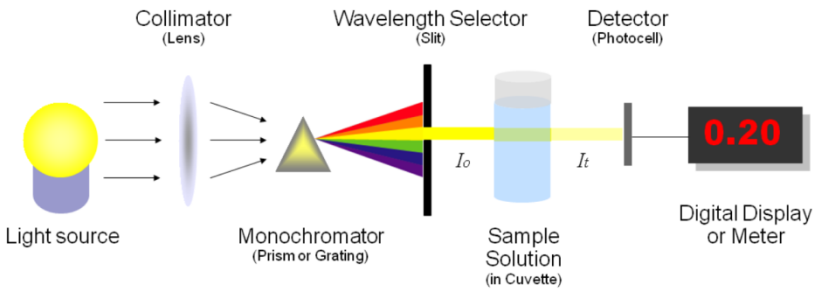


Figure 1: Basic structure of spectrophotometers (CC BY-4.0; Heesung Shim via LibreTexts)

25. Why is the glass electrode selective to hydrogen ions? [U] / [CO2]

Ans; -

- The glass electrode is selective to hydrogen ions because its special glass membrane develops a potential difference when it interacts with the solution.
- This selectivity comes from the ability of hydrogen ions to exchange with a very thin, hydrated gel layer on the glass's surface, displacing other ions.
- The resulting potential difference is a direct function of the H^+ ion concentration

26. Write a note on isoelectric focusing and its importance in protein characterization. [U] / [CO2, CO3]

Ans;-

Isoelectric focusing (IEF) is a powerful analytical technique used primarily for the separation of proteins and other biomolecules based on their isoelectric points (pI). The isoelectric point of a protein is the pH at which the molecule carries no net electrical charge. IEF operates on the principle that proteins will migrate in an electric field until they reach a point in a pH gradient where their net charge is zero—this point is their isoelectric point

27. Which analytical method you would prefer to separate the individual components of a mixture? [U] / [CO2, CO3]

Ans;-

- Chromatography can be used to separate the individual components of a mixture
- Chromatography is an important biophysical technique that enables the separation, identification, and purification of the components of a mixture for qualitative and quantitative analysis.
- Proteins can be purified based on characteristics such as size and shape, total charge, hydrophobic groups present on the surface, and binding capacity with the stationary phase.
- Four separation techniques based on molecular characteristics and interaction type use mechanisms of ion exchange, surface adsorption, partition, and size exclusion.

28. Enumerate the applications of radioisotopes in biology. [U] / [CO4]

In medicine

1.Diagnostic imaging: Radioisotopes like Technetium-99m and Iodine-131 are used in imaging techniques to diagnose organ function and identify abnormalities.

2.Medical tracers: Administered in small doses, tracers are used to study the movement of substances in the body, such as tracking blood flow.

3.Therapy: Isotopes like cobalt-60 and iodine-131 are used in radiotherapy to kill cancer cells by damaging their DNA.

In research

4.Molecular and metabolic studies: Radioisotopes of elements like C-14 are used to trace the path of molecules through metabolic pathways, like in photosynthesis or protein synthesis.

5.Cellular and organismal tracking: Researchers use radioisotopes to label cells or molecules to track their distribution and localization within a larger organism.

In agriculture

6.Crop improvement: Radioisotopes can induce mutations to develop new strains of crops that are resistant to diseases or have higher yields.

29. Explain the application of Sr-85 in bone imaging. [U] / [CO4]

Applications of Sr-85 in bone imaging

- **Detecting metastases**: Sr-85 scans were used to find and locate bone metastases from cancers like breast, lung, and prostate.,
- **Diagnosing other bone conditions**: The technique was also used for diagnosing other bone diseases, including fractures, osteomyelitis, and Paget's disease.